

Module 1 Assignment 2

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Purpose

The goal of this assignment is to explore computational reproducibility and apply the base R coding skills we've learned and practiced in class and lab.

Task

Write R code to successfully answer each question below and/or write text to successfully answer the question.

Criteria for Success

- Code is within the provided code chunks
- Code chunks run without errors
- Code produces the correct result
 - Code attempts will clear logical direction will get half points
 - Code that produces the correct answer will receive full points
- Text answers correctly address the question asked

Due Date

Sept 21 at 1 pm MST

Assignment Questions

This assignment is worth 20 points total. Each question is worth 1 point.

Definitions (1 point each)

In your own words, define/describe the following terms.

These don't need to be technical descriptions; think about how you might explain them to a friend who is not in this class.

Grading note: I've copied these definitions from my slides. If their definitions are exactly the same, they have copied directly from the slides and not used their own words, so take 0.5 points off.

1. *Reproducibility*: ability to repeat the original study using the same data, materials, and methods
2. *Open science*: Scientific research conducted and communicated in an honest, accessible, and transparent way, such that—at a minimum—a study can be reproduced and/or replicated.
3. *R*: R refers to both the programming language and the software that interprets scripts written in the language.

4. *RStudio*: RStudio is an integrated development environment (IDE) that helps us interact with R more easily.

Vectors (1 point each)

5. Assign the number 120 to an object called `birds`.

```
birds <- 120
birds
```

```
## [1] 120
```

6. Divide the object `birds` by 5. Save the result as an object called `fewer_birds`.

Remember to type the object `fewer_birds` again on the second line of code so the result is printed out when you run the entire code chunk.

```
fewer_birds <- birds / 5
fewer_birds
```

```
## [1] 24
```

7. Create a vector called `snails` with the following values: 7, 2, 19, 4, 0, 2

```
snails <- c(7, 2, 19, 4, 0, 2)
snails
```

```
## [1] 7 2 19 4 0 2
```

8. Write a line of code that uses a function to determine what *class* (type) of data is in the `snails` vector.

```
class(snails)
```

```
## [1] "numeric"
```

9. Run the line of code below. Based on the output, explain what the line of code does (~ 1 sentence).

```
snails[3:5]
```

```
## [1] 19 4 0
```

Answer: Returns the third, fourth, and fifth elements from the `snails` vector

10. Create a character vector called `rhymes` that contains the following values: `cat`, `rat`, `bat`, `hat`

```
rhymes <- c("cat", "rat", "bat", "hat")
rhymes
```

```
## [1] "cat" "rat" "bat" "hat"
```

Cactus Data Vectors

For the following questions, we will be using the cactus pad data that we collected a few weeks ago. Run the following code chunk to bring (“read”) the .csv file into R for us to use.

```
cactus <- read.csv("../data_raw/CactusData.csv")
```

This code chunk uses the `$` operator to subset individuals columns from 2D data and treat them as vectors.

Here, I’m saving the cactus pad length column as a vector called `pad_length`. Once you run this code chunk, you should see `pad_length` vector in your environment.

```
# run this code chunk
pad_length <- cactus$length_in
pad_length
```

```
## [1] 12.500 12.300 14.500 10.500 8.500 14.000 12.000 9.000 15.000 12.800
## [11] 11.000 10.500 10.750 13.000 14.000 11.500 11.500 10.000 12.000 11.000
## [21] 7.500 6.500 4.500 4.500 4.000 4.000 2.500 2.500 5.000 7.500
## [31] 13.000 14.500 10.500 12.000 13.000 14.750 10.750 11.000 14.250 10.750
## [41] 8.500 7.500 10.000 12.000 8.500 9.500 10.000 7.500 3.500 8.000
## [51] 9.000 12.700 6.500 8.750 7.500 10.500 11.000 10.570 9.500 8.000
## [61] 14.100 10.500 11.300 10.500 11.800 8.500 8.200 11.100 13.600 9.200
## [71] 9.500 5.500 6.500 8.250 6.500 7.900 6.100 6.200 5.900 1.000
## [81] 9.000 14.000 14.000 8.000 14.500 12.000 8.000 11.000 11.500 5.500
## [91] 7.125 11.125 8.500 9.500 9.125 9.000 11.000 10.250 10.250 10.750
## [101] 11.500 10.500 11.000 8.000 9.000 11.000 9.000 10.500 10.500 8.000
## [111] 16.000 13.500 15.000 12.000 8.500 11.000 13.000 3.500 11.000 10.000
```

In the next code chunk, I'm doing the same thing with a couple more columns.

```
# run this code chunk
pad_width <- cactus$width_in
damage <- cactus$damage
```

```
pad_width
```

```
## [1] 8.500 8.000 11.000 7.000 6.500 13.000 14.000 7.500 11.700 10.000
## [11] 8.000 7.000 8.000 9.000 9.000 9.000 9.500 6.500 7.500 9.500
## [21] 6.000 5.000 4.000 3.750 4.500 5.000 2.500 3.000 4.500 6.000
## [31] 11.250 10.600 7.000 8.500 9.000 12.500 7.000 7.750 10.000 6.500
## [41] 4.500 4.500 5.000 6.500 4.500 5.000 6.500 4.000 3.000 3.000
## [51] 4.800 8.600 3.200 5.500 4.500 4.750 8.900 6.250 7.000 3.900
## [61] 8.000 6.400 5.000 6.900 7.500 4.400 4.000 7.900 7.100 4.300
## [71] 5.000 2.000 2.500 3.500 2.600 3.600 3.000 3.000 3.000 0.250
## [81] 11.000 8.000 6.500 5.500 9.000 6.500 7.500 6.250 8.000 8.000
## [91] 5.000 7.250 5.000 6.000 7.125 4.500 6.500 6.850 7.000 5.500
## [101] 6.000 6.500 6.000 4.000 4.500 8.000 6.000 7.000 6.500 5.500
## [111] 6.000 7.000 9.000 8.000 4.000 11.500 6.000 1.500 8.500 9.000
```

```
damage
```

```
## [1] "Some" "Some" "Some" "Some" "Some" "Some" "Some" "Some" "Some" "Some" "Some"
## [11] "None" "None" "None" "None" "None" "None" "None" "None" "None" "None" "None"
## [21] "Some" "Most" "Some" "Some" "Some" "Some" "Some" "Some" "Some" "Some" "Some"
## [31] "Some" "Some" "Some" "None" "All" "Some" "None" "None" "None" "Most" "None"
## [41] "Some" "None" "Most" "Most" "None" "None" "None" "Most" "Most" "None"
## [51] "None" "Some" "Some" "Some" "Some" "Most" "Some" "Some" "Some" "Some" "Some"
## [61] "Some" "None" "None" "Some" "Most" "Some" "None" "Some" "None" "Some"
## [71] "Some" "Some" "Some" "Some" "Some" "Some" "Some" "Some" "None" "None"
## [81] "All" "All" "None" "Some" "None" "None" "Most" "None" "None" "None"
## [91] "None" "Some" "None" "None" "None" "Some" "None" "Some" "Some" "Some"
## [101] "None" "Some" "None" "None" "None" "None" "None" "Some" "None" "None"
## [111] "None" "None" "Most" "Some" "None" "Most" "All" "None" "Some" "All"
```

You should now see two more vectors in the Environment: `pad_width` and `damage`.

11. Write a line of code that returns the 4th value in the `pad_width` vector.

```
pad_width[4]
```

```
## [1] 7
```

12. Write a line of code that saves the 42nd value in the `pad_width` vector as an object called `answer`.

Again, remember to type the name of the object (`answer`) on the second line of code so it is printed out when you run the entire code chunk.

```
answer <- pad_width[42]
answer
```

```
## [1] 4.5
```

13. Write a line of code that pulls out the 6th through the 10th value in the `pad_width` vector.

```
pad_width[6:10]
```

```
## [1] 13.0 14.0 7.5 11.7 10.0
```

14. Using the `length()` function, find the number of elements in the `damage` vector.

```
length(damage)
```

```
## [1] 120
```

15. Write a line of code that returns all of the elements of the `damage` vector which have no damage on the pad.

```
damage[damage == "None"]
```

```
## [1] "None" "None" "None" "None" "None" "None" "None" "None" "None" "None"
## [11] "None" "None" "None" "None" "None" "None" "None" "None" "None" "None"
## [21] "None" "None" "None" "None" "None" "None" "None" "None" "None" "None"
## [31] "None" "None" "None" "None" "None" "None" "None" "None" "None" "None"
## [41] "None" "None" "None" "None" "None" "None" "None" "None" "None" "None"
```

16. Combining your answers for Questions 14 and 15, write a line of code that determines how many recorded cactus pads had no damage to them.

```
length(damage[damage == "None"])
```

```
## [1] 49
```

17. Write a line of code that returns all of the elements in the `pad_length` vector which are greater than 10.

```
pad_length[pad_length > 10]
```

```
## [1] 12.500 12.300 14.500 10.500 14.000 12.000 15.000 12.800 11.000 10.500
## [11] 10.750 13.000 14.000 11.500 11.500 12.000 11.000 13.000 14.500 10.500
## [21] 12.000 13.000 14.750 10.750 11.000 14.250 10.750 12.000 12.700 10.500
## [31] 11.000 10.570 14.100 10.500 11.300 10.500 11.800 11.100 13.600 14.000
## [41] 14.000 14.500 12.000 11.000 11.500 11.125 11.000 10.250 10.250 10.750
## [51] 11.500 10.500 11.000 11.000 10.500 10.500 16.000 13.500 15.000 12.000
## [61] 11.000 13.000 11.000
```

18. Calculate both the average (using the `mean()`) and the standard deviation (`sd()`) for all of the cactus pad lengths.

```
mean(pad_length)
```

```
## [1] 9.809542
```

```
sd(pad_length)
```

```
## [1] 3.007307
```

19. Calculate both the average (`mean()`) and the standard deviation (`sd()`) for the cactus pad widths.

```
mean(pad_width)
```

```
## [1] 6.465625
```

```
sd(pad_width)
```

```
## [1] 2.52099
```

20. Compare the results of Questions 18 and 19. Are cactus lengths or widths larger, on average? Which of the two sets of data has greater variability, and what does that mean? (~2 sentences)

Answer: Cactus pad lengths are longer, on average, than cactus pad widths. The lengths also have more variability in their values (higher st. deviation), meaning that the lengths are further away from the mean than the widths are.

Turning in Your Assignment

Follow these steps to successfully turn in your assignment on D2L.

1. Click the **Knit** button up near the top of this document. This should produce a PDF file that shows up in the **Files** panel on the bottom-right of your screen.
2. Click the empty box to the left of the PDF file.
3. Click on the blue gear near the top of the **Files** panel and choose Export.
4. Put your last name at the front of the file name when prompted, then click the Download button. The PDF file of your assignment is now in your “Downloads” folder on your device.
5. Head over to D2L and navigate to Module 1 Assignment 2. Submit the PDF file that you just downloaded.